

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12397902
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Fujishima; Yuuya

Control device of electric drive system, electric drive system and electric aircraft

Abstract

A control device of an electric drive system for rotationally driving one of a rotary blade and a screw that are a rotating body. The control device has an angle acquisition unit configured to acquire an angle of the rotating body, and a lock control unit configured to lock a rotation of the rotating body within a set lock angle range based on the acquired angle by the angle acquisition unit.

Inventors:	Fujishima; Yuuya (Kariya, JP)
Applicant:	DENSO CORPORATION (Kariya, JP)
Family ID:	1000008779816
Assignee:	DENSO CORPORATION (Kariya, JP)
Appl. No.:	17/730660
Filed:	April 27, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220258869 A1	Aug. 18, 2022

Foreign Application Priority Data

JP	2019-210176	Nov. 21, 2019
----	-------------	---------------

Related U.S. Application Data

continuation parent-doc WO PCT/JP2020/041845 20201110 PENDING child-doc US 17730660

Publication Classification

Int. Cl.: **B64U50/19** (20230101); **B64C29/00** (20060101); **B64D27/24** (20240101); **B64D31/06** (20240101); **B64U10/20** (20230101); **B64U30/297** (20230101); **B64U50/13** (20230101)

U.S. Cl.:

CPC **B64D27/24** (20130101); **B64D31/06** (20130101); **B64U10/20** (20230101); **B64U50/13** (20230101); **B64U50/19** (20230101); B64C29/0033 (20130101); B64U30/297 (20230101)

Field of Classification Search

CPC: B64D (27/24); B64D (31/06); B64U (10/20); B64U (50/13); B64U (50/19); B64U (30/297); B64U (50/14); B64C (29/0033); B64C (11/46); B64C (29/00)

USPC: 440/49

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1491512	12/1923	Bracco	440/49	B63H 1/22
5994962	12/1998	Watanabe et al.	N/A	N/A
6288509	12/2000	Amagasa	N/A	N/A
6622962	12/2002	White	244/12.3	B64C 27/26
10053213	12/2017	Tu	N/A	N/A
2004/0112053	12/2003	Yamada	60/602	F02B 39/16
2006/0125439	12/2005	Ajima et al.	N/A	N/A
2009/0119865	12/2008	Amagasa et al.	N/A	N/A
2010/0286870	12/2009	Endo et al.	N/A	N/A
2014/0225538	12/2013	Omata	318/400.02	H02P 21/34
2016/0094168	12/2015	Maekawa et al.	N/A	N/A
2016/0116304	12/2015	Iwamoto et al.	N/A	N/A
2016/0207625	12/2015	Judas et al.	N/A	N/A
2016/0244157	12/2015	Welsh	N/A	N/A
2017/0101175	12/2016	Welsh	N/A	N/A
2017/0137142	12/2016	Firanski et al.	N/A	N/A
2017/0166302	12/2016	Shiosaki	N/A	N/A
2018/0086445	12/2017	Combs	N/A	B64C 3/44
2018/0229606	12/2017	Vondrell et al.	N/A	N/A
2019/0127056	12/2018	Weekes et al.	N/A	N/A
2019/0127061	12/2018	McLaren et al.	N/A	N/A
2019/0340937	12/2018	Villa	N/A	G05D 1/0202
2020/0010184	12/2019	Welsh	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
3623288	12/2019	EP	B64C 11/00
H11-267791	12/1998	JP	N/A
2011-162173	12/2010	JP	N/A
5909939	12/2015	JP	N/A
WO-2020006659	12/2019	WO	H02K 29/08

OTHER PUBLICATIONS

SRPaero, “Replacing a VTOL Propeller”, srp.aero, 2019 (Year: 2019). cited by examiner
CTIA Research Team, “Up, Up and Away: How Do Drones Works?”, www.ctia.org (Year: 2019).
cited by examiner

Mikets, “How to tie two motors together using encoders in TELEOP?” First Tech Challenge,
ftc.forum.firstinspires.org. 2016 (Year: 2016). cited by examiner

Sony, What is the purpose for Last Known Good Configuration function and how to enable it?
(Year: 2018). cited by examiner

“Brushless DC Gear Motor Stopping Accuracy”, Feb. 4, 2018,
<https://web.archive.org/web/20180204142934/>, Retrieved Mar. 4, 2025. cited by applicant

Primary Examiner: Young; Tiffany P

Assistant Examiner: Chou; Shien Ming

Attorney, Agent or Firm: Oliff PLC

Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of International Patent Application No. PCT/JP2020/041845 filed on Nov. 10, 2020, which designated the U.S. and based on and claims the benefits of priority of Japanese Patent Application No. 2019-210176 filed on Nov. 21, 2019. The entire disclosure of all of the above applications is incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure relates to a control device of an electric drive system.

BACKGROUND

(2) In recent years, a development of manned or unmanned electric aircraft called electric vertical take-off and landing aircraft (eVTOL) has become active as a type of aircraft different from airplanes having a gas turbine engine.

SUMMARY

(3) A technique capable of locking a rotary blade or a screw as a rotating body in a specific angle range is desired.

(4) The present disclosure can be realized as the following embodiments.

(5) According to one embodiment of the present disclosure, a control device of an electric drive system is provided. This control device is a control device of an electric drive system for rotationally driving one of the rotary blades and the screw, and includes a lock control unit configured to lock a rotation of the rotating body within a set lock angle range.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) The above and other objects, features and advantages of the present disclosure will become more apparent from the following detailed description made with reference to the accompanying drawings. In the drawings:
- (2) FIG. 1 is a perspective view showing an external configuration of an electric vertical take-off and landing aircraft equipped with a control device of an electric drive system;
- (3) FIG. 2 is a block diagram showing a schematic configuration of an electric vertical take-off and landing aircraft;
- (4) FIG. 3 is a block diagram for explaining a schematic configuration of a control device;
- (5) FIG. 4 is an explanatory diagram for explaining a target lock position;
- (6) FIG. 5 is an explanatory diagram for explaining a method of locking at a reference angle as a target;
- (7) FIG. 6 is a flowchart showing a procedure of lock processing;
- (8) FIG. 7 is a schematic diagram showing a connection relationship between a motor and a rotating body in a second embodiment;
- (9) FIG. 8 is an explanatory diagram for explaining a method of locking at a reference angle as a target;
- (10) FIG. 9 is an explanatory diagram showing one example of a relationship between a rotation angle and an electric angle;
- (11) FIG. 10 is an explanatory diagram showing another example of the relationship between the rotation angle and the electric angle;
- (12) FIG. 11 is a flowchart showing a procedure of lock processing in the second embodiment;
- (13) FIG. 12 is a flowchart showing a procedure of lock processing in a third embodiment;
- (14) FIG. 13 is a flowchart showing a procedure of lock processing in a fourth embodiment;
- (15) FIG. 14 is a flowchart showing a procedure of lock processing in a fifth embodiment;
- (16) FIG. 15 is a flowchart showing a procedure of lock processing in a sixth embodiment;
- (17) FIG. 16 is a perspective view showing a schematic configuration of an electric vertical take-off and landing vehicle according to a seventh embodiment;
- (18) FIG. 17 is a flowchart showing a procedure of lock processing in the seventh embodiment; and
- (19) FIG. 18 is an explanatory diagram showing how a pulse is output at a target lock position.

DETAILED DESCRIPTION

(20) In recent years, a development of manned or unmanned electric aircraft called electric vertical take-off and landing aircraft (eVTOL) has become active as a type of aircraft different from airplanes having a gas turbine engine. The electric vertical take-off and landing aircraft is equipped with a plurality of electric drive systems (EDS) for rotationally driving the rotary blades, and the lift and thrust of an airframe can be obtained by rotationally driving the multiple rotary blades by multiple motors.

(21) In the electric vertical take-off and landing aircraft, some rotary blades may not be used and may not rotate depending on flight conditions and the like. A discloser of the present application has found that there are situations in which a non-rotating rotor is desired to be locked in a specific angle range in order to achieve a predetermined purpose. The discloser of the present application assumes for the predetermined purposes that, for example, to accommodate a non-rotating rotary blade in an accommodating portion provided in the electric vertical take-off and landing aircraft, to suppress a deterioration of efficiency due to an air resistance of the non-rotating rotary blades, and to suppress an occurrence of problems caused by an idle rotation of the rotary blades, etc. The above-mentioned situations are common not only in the electric drive system mounted on the electric aircraft such as an electric vertical take-off and landing machine, but also in an electric

drive system mounted on a ship or the like for rotationally driving a screw. Therefore, a technique capable of locking a rotary blade or a screw as a rotating body in the specific angle range is desired. (22) The present disclosure can be realized as the following embodiments.

(23) According to one embodiment of the present disclosure, a control device of an electric drive system is provided. This control device is a control device of an electric drive system for rotationally driving one of the rotary blades and the screw, and includes a lock control unit configured to lock a rotation of the rotating body within a set lock angle range.

(24) According to the control device of the electric drive system of this embodiment, since the lock control unit for locking the rotation of the rotating body within the set lock angle range is provided, the rotating body can be locked within the lock angle range. Therefore, for example, a rotating body that is not rotationally driven can be accommodated in an accommodating portion, and deterioration of efficiency due to resistance that the rotating body that is not rotationally driven receives from an external fluid can be suppressed.

(25) The present disclosure can be realized as the following embodiments. For example, the present disclosure can be realized in the embodiment of a control method of an electric drive system, an electric aircraft provided with a control device of the electric drive system, an electric moving body, or the like.

A. First Embodiment

A-1. Device Configuration

(26) A control device as one embodiment of the present disclosure shown in FIG. 1 is mounted on an electric drive system **10** (hereinafter, also referred to as “EDS (Electric Drive System) **10**”) to control an operation of the EDS **10**. In the present embodiment, a plurality of EDS **10**s are mounted on an electric vertical take-off and landing aircraft **100** (hereinafter, also referred to as “eVTOL **100**”). The eVTOL **100** is configured as a manned aircraft that is electrically driven and can take off and land in a vertical direction. The eVTOL **100** includes an airframe **20**, a plurality of rotary blades **30**, and a plurality of EDSs **10** provided corresponding to each rotary blade **30** for rotationally driving each rotary blade **30**. The eVTOL **100** of the present embodiment includes six rotary blades **30** and six EDSs **10**, respectively. The number of rotary blades **30** and EDSs **10** included in the eVTOL **100** is not limited to six, and may be any number such as four or nine.

(27) The airframe **20** corresponds to a portion of the eVTOL **100** excluding the six rotary blades **30** and the six EDSs **10**. The airframe **20** includes an airframe main body **21**, two main wings **25**, and a tail wing **28**.

(28) The airframe main body **21** constitutes a body portion of the eVTOL **100** and is formed along a front-rear direction of the eVTOL **100**. A passenger compartment **22** is formed inside the airframe main body **21** on a front side. Two rotary blades **30** are arranged side by side in the front-rear direction on a vertically upper surface on a rear side of the airframe main body **21**. The two main wings **25** are formed so as to extend from the airframe main body **21** in a right direction and a left direction, respectively. Rotary blade **30** is arranged substantially in a center of each main wing **25**. A rotary blade **30** is arranged at a tip of each main wing **25**. The tail wing **28** is formed at a rear end portion of the airframe main body **21**.

(29) Of the six rotary blades **30**, the two rotary blades **30** arranged in the aircraft main body **21** and the two rotary blades **30** arranged in the substantially central portion of each main wing **25** are mainly configured as the lift rotary blades **31** for obtaining the lift of the airframe **20**. The lift rotary blade **31** is rotationally driven when an operation mode of the eVTOL **100** is the takeoff mode or the like. Further, the lift rotary blade **31** may not be rotationally driven when the operation mode of the eVTOL **100** is a cruising mode or the like. In the present embodiment, each lift rotary blade **31** is locked in a specific angle range in a state where it is not rotationally driven, and is housed in an accommodating portion **35** formed vertically below each lift rotary blade **31**. The accommodating portion **35** is formed along a direction substantially parallel to the front-rear direction of the eVTOL **100**. A detailed description of the lock of the lift rotary blade **31** will be described later. A

tilt angle of each lift rotary blade **31** is fixed.

(30) Of the six rotary blades **30**, the two rotary blades **30** arranged at the tip of each main wing **25** are configured as tilt rotors **32**. Each tilt rotor **32** is configured so that a tilt angle can be changed. In each tilt rotor **32**, the tilt angle is set so that a rotation axis of the tilt rotor **32** is along the vertical direction as shown in FIG. **1** when the operation mode of the eVTOL **100** is a takeoff mode. Unlike the state shown in FIG. **1**, when the operation mode of the eVTOL **100** is a cruising mode or the like, the tilt angle of each tilt rotor **32** is set so that the rotation axis of the tilt rotor **32** is along the horizontal direction. The tilt rotor **32** may not be rotationally driven when the operation mode of the eVTOL **100** is a landing mode or the like. In the present embodiment, each tilt rotor **32** is locked in a specific angle range in a state where it is not rotationally driven. A detailed description of the lock of the tilt rotor **32** will be described later.

(31) The six rotary blades **30** are rotationally driven independently of each other around their respective rotation axes. Each rotary blades **30** has blades **33** formed symmetrically about the rotation axis along a direction perpendicular to the rotation axis. The blade **33** has a substantially plate-like appearance shape, and a rotation shaft is inserted through a substantially central portion. In the present embodiment, each rotary blade **30** has two blades **33**, respectively. The two blades **33** are configured to be independently rotatable.

(32) The six EDSs **10** are configured as drive devices for rotationally driving each rotary blade **30**. Four of the six EDSs **10** rotate and drive the lift rotary blade **31** respectively. Two of the six EDSs **10** rotate the tilt rotor **32**, respectively. The configurations of each EDS **10** are approximately equal to each other.

(33) As shown in FIG. **2**, each EDS **10** includes a motor **11**, an inverter circuit **12**, and a control device **50**.

(34) The motor **11** outputs torque corresponding to the voltage and current supplied from the inverter circuit **12**, and drives the rotary blade **30** to rotate via the shaft **19**. A rotation control of the motor **11** is executed so that a torque as instructed is output using an electric angle. In the present embodiment, the motor **11** is composed of a brushless motor, but is not limited to the brushless motor, and may be configured by any motor such as an induction motor or a reluctance motor. In the present embodiment, the output of the motor **11** is transmitted to the rotary blade **30** without going through a gear. Therefore, a mechanical angle of the motor **11** and a rotation angle of the rotary blade **30** match. As shown in FIG. **3**, the motor **11** of the present embodiment is provided with an angle sensor **13** configured by a resolver or the like. The angle sensor **13** may be omitted.

(35) The inverter circuit **12** shown in FIG. **2** includes switching elements such as an IGBT (Insulated Gate Bipolar Transistor) and a MOSFET (Metal-Oxide-Semiconductor Field-Effect Transistor), and converts a DC voltage supplied from a power supply unit **80** into a three-phase AC voltage and supplies it to the motor **11**. The inverter circuit **12** switches a switching element at a duty ratio according to a control signal supplied from the control device **50**.

(36) The control device **50** shown in FIG. **3** is composed of a computer having a CPU (Central Processing Unit), a ROM (Read Only Memory), and a RAM (Random Access Memory). The CPU functions as an inverter control unit **51**, a lock control unit **52**, and an angle acquisition unit **53** by executing a control program stored in advance in the ROM. The ROM and RAM of the control device **50** also function as a reference angle storage unit **54**. Further, the control device **50** has a pulse output unit **55**.

(37) The inverter control unit **51** controls the operation of the inverter circuit **12** by outputting a control signal to the inverter circuit **12**. The lock control unit **52** outputs a control signal to the inverter control unit **51** to control the motor **11**, so that the rotation of the rotary blade **30** is locked within a preset angle range (hereinafter, also referred to as "lock angle range"). The angle acquisition unit **53** acquires a current angle of the motor **11**. In the present embodiment, the angle acquisition unit **53** acquires the electric angle of the motor **11** by using the value measured by the angle sensor **13**, but the electric angle of the motor **11** may be estimated from a value measured by

a current sensor (not shown) provided in each phase of the inverter circuit **12**. The reference angle storage unit **54** stores a reference angle. The reference angle means an angle corresponding to a target lock position when the rotation of the rotary blade **30** is locked within the lock angle range. The reference angle of the present embodiment is represented by the electric angle of the motor **11** as described later. The pulse output unit **55** is composed of a circuit including a transistor, and outputs a pulse according to a control signal received from the lock control unit **52**. A detailed description of the lock control unit **52**, the angle acquisition unit **53**, the reference angle storage unit **54**, and the pulse output unit **55** will be described later.

(38) As shown in FIG. 2, various components for controlling each EDS **10** are arranged in the eVTOL **100**. Specifically, the eVTOL **100** includes a flight control system **40**, a communication device **70**, the power supply unit **80**, and a user interface unit **90** (hereinafter, also referred to as “UI unit **90**”).

(39) The flight control system **40** is composed of a computer having a CPU, a ROM and a RAM. The CPU controls the overall operation of the eVTOL **100** by executing a control program stored in the ROM in advance. The overall operation of the eVTOL **100** corresponds to, for example, a takeoff operation, a landing operation, a cruising operation, and the like. Such an operation may be performed based on a preset flight program or may be performed by the maneuvering of an occupant. The flight control system **40** is connected to each EDS **10**, and outputs a control command to the control device **50** of each EDS **10** according to the operation mode of the eVTOL **100**. Further, the flight control system **40** outputs a lock command for stopping and locking the rotation of the rotary blade **30** that is not driven to rotate to the control device **50** of each EDS **10**. For example, when the operation mode of the eVTOL **100** is the cruising mode, the lock command is output to the control device **50** of the EDS **10** corresponding to the lift rotary blade **31**. Further, for example, when the operation mode of the eVTOL **100** is the landing mode, the lock command is output to the control device **50** of the EDS **10** corresponding to the tilt rotor **32**. A detailed description of the lock of the rotary blade **30** will be described later.

(40) The communication device **70** communicates with another eVTOL **100**, a control tower on the ground, and the like. The communication device **70** corresponds to, for example, a civilian VHF (Very High Frequency) radio. In addition to the civilian VHF, the communication device **70** may be configured as a device that performs communication such as a wireless LAN specified in IEEE802.11 or a wired LAN specified in IEEE802.3.

(41) The power supply unit **80** functions as one of the power supply sources in the eVTOL **100**, and supplies three-phase AC power to the motor **11** via the inverter circuit **12** of each EDS **10**. In the present embodiment, the power supply unit **80** is composed of a lithium ion battery, but is not limited to the lithium ion battery and may be composed of any secondary battery such as a nickel hydrogen battery, and may be an arbitrary power supply source such as a fuel cell or a generator in place of or in addition to the secondary battery.

(42) The UI unit **90** supplies a predetermined user interface. The user interface includes, for example, an operation input unit such as a keyboard and a button, a display unit such as a liquid crystal panel, and the like. The UI unit **90** is provided in, for example, the passenger compartment **22** of the eVTOL **100**. The occupant can change the operation mode of the eVTOL **100** by using the UI unit **90**.

(43) The control device **50** of the EDS **10** in the present embodiment executes the lock process described later, so that the control device **50** locks the rotary blade **30** in the currently rotationally driven state by stopping the rotation of the rotary blade **30** at the target lock position as a target when a non-rotary driven state is planned. The lock angle range described above means a range allowed as a range of deviation with respect to the target lock position. Therefore, the lock angle range includes the target lock position.

(44) A relationship between the target lock position, and the mechanical angle and the electric angle of the motor **11** will be described with reference to FIGS. 4 and 5. In FIG. 4, the mechanical angle

of the motor **11** is shown by an alternate long and short dash line, the electric angle is shown by a thin solid line, and a resolver angle is shown by a thick solid line as a reference. Further, a horizontal axis of FIG. 5 indicates a time (s). In the present embodiment, since the output of the motor **11** is transmitted to the rotary blade **30** without going through the gear, the target lock position of the rotary blade **30** can be represented by the mechanical angle of the motor **11**. In the examples shown in FIGS. 4 and 5, the lock angle range is not shown.

(45) As described above, the rotation control of the motor **11** is performed using the electric angle. In the examples shown in FIGS. 4 and 5, the mechanical angle (deg) and the electric angle (deg) in the motor **11** having 4 pole pairs are shown. The number of pole pairs corresponds to the number of magnetic poles of the motor **11** divided by two. The number of cycles of the electric angle per unit time corresponds to the value obtained by multiplying the number of cycles of the mechanical angle per unit time by the number of pole pairs. In the examples shown in FIGS. 4 and 5, it is established that a relationship of the number of cycles of the mechanical angle of the motor **11** per unit time and the number of cycles of the electric angle of the motor **11** per unit time is equal to a relationship between 1 and 4. During one cycle in which the motor **11** makes one rotation and the mechanical angle is from 0 deg to 360 deg, there are four cycles of electric angles from 0 deg to 360 deg.

A-2. Lock Processing

(46) The lock processing shown in FIG. 6 is executed in the control device **50** of each EDS **10** when the rotation of the motor **11** is started in each EDS **10**.

(47) The lock control unit **52** detects a reference angle, that is, an angle corresponding to a target lock position when the rotary blade **30** is locked in the lock angle range (step **S110**). In the present embodiment, the reference angle is stored in the reference angle storage unit **54** as the value of the electric angle of the motor **11** when the rotation of the rotary blade **30** is locked last time.

Therefore, in step **S110**, the value of the electric angle is read out from the reference angle storage unit **54** and detected. The reference angle may be stored in the reference angle storage unit **54** as the value of the electric angle when a start switch of the eVTOL **100** is turned on (time **t0**). In FIG. 5, a time when the rotation of the motor **11** starts is shown as **t0**. As described above, when the value of the electric angle at the time of the previous lock is the reference angle and the eVTOL **100** is stopped in the locked state, the value of the electric angle when the start switch of the eVTOL **100** is turned on (time **t0**) next time corresponds to the reference angle.

(48) As shown in FIG. 6, the lock control unit **52** counts up the number of times the electric angle of the motor **11** reaches 0 deg (step **S120**). In the example shown in FIG. 5, the number of counters is one at time **t1**, two at time **t2**, three at time **t3**, and four at time **t4** as the motor **11** rotates.

(49) As shown in FIG. 6, the lock control unit **52** specifies whether or not the number of counters has reached the number of pole pairs (step **S130**). When it is specified that the number of times of the counter has not reached the number of pole pairs (step **S130**: NO), the process returns to step **S120**. On the other hand, when it is specified that the number of counters has reached the number of pole pairs (step **S130**: YES), the lock control unit **52** specifies whether or not a current electric angle of the motor **11** has reached the reference angle (step **S140**). As shown in FIG. 5, the electric angle at time **t4** when the number of counters reaches the number of pole pairs is 0 deg. The value of the electric angle increases with the rotation of the motor **11** and reaches the reference angle at time **t5**.

(50) As shown in FIG. 6, when it is specified that the current electric angle of the motor **11** has not reached the reference angle (step **S140**: NO), the lock control unit **52** repeats step **S140**. On the other hand, when it is specified that the current electric angle of the motor **11** has reached the reference angle (step **S140**: YES), the lock control unit **52** causes the pulse output unit **55** to output a pulse (step **S150**). In the example shown in FIG. 5, the pulse is output at time **t5**. By being controlled in this way, the pulse output unit **55** outputs a pulse each time the rotational position of the rotary blade **30** reaches the target lock position.

(51) As shown in FIG. 6, the lock control unit 52 resets the number of counters (step S160). The lock control unit 52 determines whether or not there is a lock command from the flight control system 40 (step S170). If it is determined that there is no lock command (step S170: NO), the process returns to step S120. On the other hand, when it is determined that there is the lock command (step S170: YES), the rotation of the rotary blade 30 is locked by locking the rotation of the motor 11 (step S180). In step S180, the rotation of the motor 11 is locked as a target at the reference angle first reached after the number of counters reaches the number of pole pairs of the motor 11, and the rotation of the rotary blade 30 is locked. As shown in FIG. 3, the rotation lock of the motor 11 may be controlled so that the current electric angle of the motor 11 acquired by the angle acquisition unit 53 matches the reference angle stored in the reference angle storage unit 54. At this time, instead of suddenly stopping the rotation of the motor 11, the rotation speed may be gradually reduced, and the rotation of the motor 11 may be stopped and locked when the reference angle is reached after the rotation speed becomes equal to or less than a predetermined rotation speed. Execution of step S180 ends the lock processing.

(52) In the lock processing of the present embodiment, when the operation mode of the eVTOL 100 is the cruising mode or the like, the lift rotary blade 31 that is not driven to rotate is locked at the target lock position within the lock angle range. Here, the lock angle range of the lift rotary blade 31 is included in an accommodation angle range which is an angle range in which the lift rotary blade 31 can be housed in the accommodating portion 35. Therefore, the lift rotary blade 31 is stopped and locked at an angle at which the lift rotary blade 31 can be housed in the accommodating portion 35. As described above, the lift rotary blade 31 of the present embodiment has two blades 33 and is housed in the accommodating portion 35. When the two blades 33 are housed in the accommodating portion 35, the two blades 33 are overlapped so as to coincide with each other in a longitudinal direction. At this time, one of the two blades 33 may be locked at the target lock position, and the other blade 33 may be overlapped with the one blade 33. In this case, the rotary blade 30 may be provided with a mechanism for overlapping the other blade 33 with the one blade 33. Further, the two blades 33 may be controlled so as to be independently locked at their respective target lock positions.

(53) Further, in the lock processing of the present embodiment, when the operation mode of the eVTOL 100 is the landing mode or the like, the tilt rotor 32 that is not driven to rotate is locked within the lock angle range. The tilt rotor 32 is stopped and locked at an angle such that the value of air resistance becomes small. In the present embodiment, the tilt rotor 32 is stopped from rotating at an angle at which the air resistance value received by the tilt rotor 32 according to the lock position of the tilt rotor 32 becomes smaller than an intermediate value between a maximum value and a minimum value and is locked. More specifically, the tilt rotor 32 is stopped and locked at an angle at which the air resistance value becomes minimum.

(54) Further, in the present embodiment, the lock angle range is $[B-2\pi/A]$ or more $[B+2\pi/A]$ or less when the number of blades 33 of the rotary blade 30 is A, the target lock position is B, and 180 deg is π .

(55) According to the control device 50 of the EDS 10 in the first embodiment described above, since the lock control unit 52 that locks the rotation of the rotary blade 30 within the set lock angle range is provided, the rotary blade 30 can be locked within the lock angle range. Therefore, for example, the rotary blade 30 that is not driven to rotate can be housed in the accommodating portion 35, and the deterioration of efficiency due to the air resistance of the rotary blade 30 that is not driven to rotate can be suppressed.

(56) In the configuration in which the rotary blade 30 that is not driven to rotate is not locked, the life of the rotary blade 30 and the motor 11 may be deteriorated or a failure of the rotary blade 30 and the motor 11 may be induced due to an idle rotation of the rotary blade 30. Further, due to such a failure, the rotary blade 30 may not operate as instructed in a situation where the rotary blade 30 is desired to be rotationally driven. As described above, in the configuration in which the rotary

blade **30** that is not driven to rotate is not locked, a defect due to the idle rotation of the rotary blade **30** may occur.

(57) On the other hand, according to the control device **50** of the EDS **10** of the present embodiment, since the rotary blade **30** that is not rotationally driven is locked within the lock angle range, it is possible to suppress an occurrence of the defect caused by the idle rotation of the rotary blade **30**.

(58) Further, the lock angle range of the lift rotary blade **31** is included in an accommodation angle range which is the angle range in which the lift rotary blade **31** can be housed in the accommodating portion **35**. Therefore, the lift rotary blade **31** that is not rotationally driven can be easily housed in the accommodating portion **35** provided in the airframe **20** of the eVTOL **100**. Further, since the lock angle range is included in the accommodation angle range, it is possible to omit providing a guide or the like in the accommodating portion **35** for accommodating the lift rotary blade **31**, and it is possible to prevent the configuration of the accommodating portion **35** from becoming complicated.

(59) Further, the lock angle range in the tilt rotor **32** is an angle range in which the air resistance value is smaller than an intermediate value between a maximum value and a minimum value of the air resistance value received by the tilt rotor **32** according to the lock position of the tilt rotor **32**. Therefore, it is possible to effectively suppress the deterioration of efficiency due to the air resistance of the tilt rotor **32** that is not driven to rotate. Further, since the tilt rotor **32** is locked by stopping the rotation at an angle that minimizes the air resistance value, deterioration of efficiency can be further suppressed.

(60) Further, the lock angle range is $[B-2\pi/A]$ or more $[B+2\pi/A]$ or less when the number of blades **33** of the rotary blade **30** is A , the target lock position is B , and 180 deg is π . Therefore, it is possible to prevent the angle at which the rotary blade **30** is locked from being significantly deviated from the target lock position.

(61) Further, since the control device **50** has the reference angle storage unit **54** that stores the reference angle for locking the rotary blade **30** in the lock angle range as the value of the electric angle of the motor **11**, the motor **11** can be locked using the reference angle. It is possible to prevent the control for locking the rotation of the motor **11** from becoming complicated.

(62) Further, the lock control unit **52** counts up the number of times the electric angle of the motor **11** reaches 0 deg , and the rotation of the motor **11** is locked as a target at the reference angle first reached after the number of counters reaches the number of pole pairs of the motor **11**, and the rotation of the rotary blade **30** is locked. Therefore, since the rotation of the rotary blade **30** can be locked by using the electric angle of the motor **11**, a detection unit for detecting the current rotational position of the rotary blade **30** and an angle sensor and the like for detecting the current rotation angle of the rotary blade **30** can be omitted, and further, an angle sensor **13** of the motor can be omitted. Therefore, it is possible to suppress the complication of the structure of the eVTOL **100**, the rotary blade **30**, and the EDS **10**, and it is possible to suppress an increase in the cost required for manufacturing the eVTOL **100**, the rotary blade **30**, and the EDS **10**.

B. Second Embodiment

(63) As shown in FIG. 7, in the EDS **10** on which the control device **50** of a second embodiment is mounted, the output of the motor **11** is transmitted to the rotary blade **30** via the gear G , and in this respect, the EDS **10** of the second embodiment is different from the EDS **10** of the first embodiment. Along with this difference, the lock processing of the second embodiment is different from the lock processing of the first embodiment in that step $S130a$ is executed instead of step $S130$. Since the other configurations are the same as those in the first embodiment, the same configurations are designated by the same reference numerals, and detailed description thereof will be omitted.

(64) In the EDS **10** of the second embodiment, the output of the motor **11** is transmitted to the rotary blade **30** via the gear G . Therefore, the mechanical angle of the motor **11** and the rotation

angle of the rotary blade **30** do not match. Therefore, in order to stop the rotation of the rotary blade **30** and lock the rotary blade **30** at the target lock position, it is necessary to estimate the rotation angle of the rotary blade **30** from the electric angle of the motor **11** in consideration of a gear ratio.

(65) FIGS. **8** to **10** show a relationship between the rotation angle of the rotary blade **30** and the mechanical angle of the motor **11** and the electric angle of the motor **11**. FIG. **8** shows a case where a gear ratio of the gear **G** is 2 and the number of pole pairs of the motor **11** is 4. Since the gear ratio of the gear **G** is 2, the rotary blade **30** makes two rotations while the motor **11** makes one rotation. Therefore, there are two cycles of electric angles from 0 deg to 360 deg during the period when the rotary blade **30** makes one rotation and the mechanical angle is from 0 deg to 180 deg. The ratio of the rotation angle of the rotary blade **30** to the electric angle of the motor **11** can be calculated by using the gear ratio of the gear **G** and the number of pole pairs of the motor **11**. In the example shown in FIG. **8**, a relationship between the number of cycles of the rotation angle of the rotary blade **30** per unit time, the number of cycles of the mechanical angle of the motor **11** per unit time, and the number of cycles of the electric angle of the motor **11** per unit time is a relationship between 2, 1, and 4. Further, in FIG. **9**, a relationship between the number of cycles of the rotation angle of the rotary blade **30** per unit time, the number of cycles of the mechanical angle of the motor **11** per certain unit time, and the number of cycles of the electric angle of the motor **11** per certain unit time is a relationship between 3, 12, and 32. In FIG. **10**, a relationship between the number of cycles of the rotation angle of the rotary blade **30** per unit time, the number of cycles of the mechanical angle of the motor **11** per unit time, and the number of cycles of the electric angle of the motor **11** per unit time is a relationship between 8, 3, and 12.

(66) The electric angle of the motor **11** at the timing when the rotation angle of the rotary blade **30** is a certain value periodically shows the same value. Therefore, the electric angle of the motor **11** when the rotation angle of the rotary blade **30** is in the target lock position periodically shows the same value. More specifically, at least each time the timing of a least common multiple of the value of the ratio of the electric angle of the motor **11** to the rotation angle of the rotary blade **30** is reached, the rotation angle of the rotary blade **30** shows the same value. The least common multiple of the value of the ratio of the electric angle of the motor **11** to the rotation angle of the rotary blade **30** is “2”, since for example, in the example shown in FIG. **8**, the relationship between the rotation angle of the rotary blade **30** and the electric angle of the motor **11** is the relationship between 2 and 4, namely 1 and 2. Therefore, the rotation angle of the rotary blade **30** and the electric angle of the motor **11** are also 0 deg for every two cycles of the electric angle of the motor **11**. Further, for example, in the example shown in FIG. **9**, since the relationship between the rotation angle of the rotary blade **30** and the electric angle of the motor **11** is the relationship between 3 and 32, the least common multiple of the ratio of the electric angle of the motor **11** to the rotation angle of the rotary blade **30** is “96”. Therefore, every 96 cycles of the electric angle of the motor **11**, the rotation angle of the rotary blade **30** and the electric angle of the motor **11** are also 0 deg. Further, for example, in the example shown in FIG. **10**, since the relationship between the rotation angle of the rotary blade **30** and the electric angle of the motor **11** is the relationship between 8 and 12, namely 2 and 3, the least common multiple of the ratio of the electric angle of the motor **11** to the rotation angle of the rotary blade **30** is “6”. Therefore, every 6 cycles of the electric angle of the motor **11**, the rotation angle of the rotary blade **30** and the electric angle of the motor **11** are also 0 deg.

(67) In the lock processing of the second embodiment shown in FIG. **11**, after the execution of step **S120**, the lock control unit **52** specifies whether the number of times of counters has reached the least common multiple of the value of the ratio of the electric angle of the motor **11** to the rotation angle of the rotary blade **30** (step **S130a**). When it is specified that the number of times of the counter does not reach the least common multiple of the value of the ratio of the electric angle of the motor **11** to the rotation angle of the rotary blade **30** (step **S130a**: NO), the process returns to step **S120**. On the other hand, when it is specified that the number of counters has reached the least

common multiple of the value of the ratio of the electric angle of the motor **11** to the rotation angle of the rotary blade **30** (step **S130a**: YES), the lock control unit **52** specifies whether or not the current electric angle of the motor **11** has reached the reference angle (step **S140**). By being controlled in this way, the pulse output unit **55** outputs a pulse each time the electric angle of the motor **11** is the reference angle and the rotational position of the rotary blade **30** reaches the target lock position. In step **S180** of the lock processing of the second embodiment, the rotation of the motor is locked as a target at the reference angle first reached after the number of counters reaches the least common multiple of the value of the ratio of the electric angle of the motor **11** to the rotation angle of the rotary blade **30**, and the rotation of the rotary blade **30** is locked. Execution of step **S180** ends the lock processing.

(68) According to the control device **50** of the EDS **10** in the second embodiment described above, the same effect as that in the first embodiment is obtained. In addition, even in a configuration in which the output of the motor **11** is transmitted to the rotary blade **30** via the gear **G**, the rotation of the rotary blade **30** can be locked within the lock angle range.

C. Third Embodiment

(69) As shown in FIG. **12**, the lock processing executed in the control device **50** of a third embodiment is different from the lock processing of the first embodiment in that steps **S112b**, **S114b** and **S140b** are executed instead of steps **S120** to **S160**. Since the other configurations are the same as those in the first embodiment, the same configurations are designated by the same reference numerals, and detailed description thereof will be omitted.

(70) After executing step **S110**, the lock control unit **52** calculates the mechanical angle corresponding to the reference angle detected in step **S110** (step **S112b**). The mechanical angle is calculated by dividing the detected reference angle by the number of pole pairs of the motor **11**.

(71) The lock control unit **52** calculates the current mechanical angle of the motor **11** from the current electric angle of the motor **11** (step **S114b**). The current mechanical angle of the motor **11** is calculated by dividing the current electrical angle of the motor **11** by the number of pole pairs of the motor **11**. The lock control unit **52** specifies whether or not the current mechanical angle of the motor **11** calculated in step **S114b** has reached the mechanical angle corresponding to the reference angle of the motor **11** calculated in step **S112b** (step **S140b**). When it is specified that the current mechanical angle of the motor **11** has not reached the mechanical angle corresponding to the reference angle (step **S140b**: NO), the process returns to step **S114b**. On the other hand, when it is specified that the current mechanical angle of the motor **11** has reached the mechanical angle corresponding to the reference angle (step **S140b**: YES), the lock control unit **52** determines whether or not there is a lock command from the flight control system **40** (step **S170**). If it is determined that there is no lock command (step **S170**: NO), the process returns to step **S114b**. On the other hand, when it is determined that there is the lock command (step **S170**: YES), the rotation of the rotary blade **30** is locked by locking the rotation of the motor **11** (step **S180**). In step **S180**, the rotation of the motor **11** is locked as a target when the current mechanical angle of the motor **11** reaches the mechanical angle of the motor **11** at the reference angle and the rotation of the rotary blade **30** is locked. Execution of step **S180** ends the lock processing.

(72) According to the control device **50** of the EDS **10** in the third embodiment described above, the same effect as that in the first embodiment is obtained. In addition, since the lock control unit **52** calculates the mechanical angle corresponding to the electric angle, it is possible to omit counting up the number of times the electric angle of the motor **11** reaches 0 deg and outputting a pulse at the reference angle.

D. Fourth Embodiment

(73) In the EDS **10** on which the control device **50** of a fourth embodiment is mounted, the output of the motor **11** is transmitted to the rotary blade **30** via the gear **G**, and in this respect, the EDS **10** of the fourth embodiment is different from the EDS **10** of the third embodiment. Along with this difference, the lock processing of the fourth embodiment shown in FIG. **13** is different from the

lock processing of the third embodiment in that steps **S112c**, **S114c** and **S140c** are executed instead of steps **S112b**, **S114b** and **S140b**. Since the other configurations are the same as those in the third embodiment, the same configurations are designated by the same reference numerals, and detailed description thereof will be omitted.

(74) After executing step **S110**, the lock control unit **52** calculates the rotation angle of the rotary blade **30** corresponding to the reference angle detected in step **S110** (step **S112c**). The rotation angle is calculated by dividing the detected reference angle by the number of pole pairs of the motor **11** and multiplying the divided value by the gear ratio of the gear **G**.

(75) The lock control unit **52** calculates the current rotation angle of the rotary blade **30** from the current electric angle of the motor **11** (step **S114c**). The current rotation angle of the rotary blade **30** is calculated by dividing the current electric angle of the motor **11** by the number of pole pairs of the motor **11** and multiplying the divided value by the gear ratio of the gear **G**. The lock control unit **52** specifies whether or not the current rotation angle of the rotary blade **30** calculated in step **S114c** has reached the rotation angle of the rotary blade **30** corresponding to the reference angle of the motor **11** calculated in step **S112c** (step **S140c**). When it is specified that the current rotation angle of the rotary blade **30** has not reached the rotation angle corresponding to the reference angle (step **S140c**: NO), the process returns to step **S114c**. On the other hand, when it is specified that the current rotation angle of the rotary blade **30** has reached the rotation angle corresponding to the reference angle (step **S140c**: YES), the lock control unit **52** determines whether or not there is a lock command from the flight control system **40** (step **S170**). If it is determined that there is no lock command (step **S170**: NO), the process returns to step **S114c**. On the other hand, when it is determined that there is the lock command (step **S170**: YES), the rotation of the rotary blade **30** is locked by locking the rotation of the motor **11** (step **S180**). In step **S180**, the rotation of the motor **11** is locked as a target when the current rotation angle of the rotary blade **30** reaches the rotation angle of the rotary blade **30** at the reference angle and the rotation of the rotary blade **30** is locked. Execution of step **S180** ends the lock processing.

(76) According to the control device **50** of the EDS **10** in the fourth embodiment described above, the same effect as that in the third embodiment is obtained. In addition, even in a configuration in which the output of the motor **11** is transmitted to the rotary blade **30** via the gear **G**, the rotation of the rotary blade **30** can be locked within the lock angle range.

E. Fifth Embodiment

(77) In the control device **50** of a fifth embodiment, the angle acquisition unit **53** acquires the mechanical angle of the motor **11** using the value measured by the angle sensor **13** provided in the motor **11**, and the reference angle storage unit **54** stores the reference angle, which is the angle corresponding to the target lock position as the value of the mechanical angle of the motor **11**. In this respect, the control device **50** of the fifth embodiment differs from the control device **50** of the third embodiment. Along with this difference, the lock processing of the fifth embodiment is different from the lock processing of the third embodiment in that steps **S112b** and **S114b** are omitted and step **S140d** is executed instead of step **S140b**. Since the other configurations are the same as those in the third embodiment, the same configurations are designated by the same reference numerals, and detailed description thereof will be omitted. In the present embodiment, the output of the motor **11** is transmitted to the rotary blade **30** without going through a gear.

(78) The lock control unit **52** detects a reference angle, that is, an angle corresponding to a target lock position when the rotary blade **30** is locked in the lock angle range (step **S110**). In the present embodiment, the reference angle is stored in the reference angle storage unit **54** as the value of the mechanical angle of the motor **11** when the rotation of the rotary blade **30** is locked last time. The reference angle may be stored in the reference angle storage unit **54** as the value of the mechanical angle when the start switch of the eVTOL **100** is turned on.

(79) The lock control unit **52** specifies whether or not the current mechanical angle of the motor **11** has reached the reference angle (step **S140d**). The current mechanical angle of the motor **11** is

measured by the angle sensor **13** provided in the motor **11** and acquired by the angle acquisition unit **53**. When it is specified that the current mechanical angle of the motor **11** has not reached the reference angle (step **S140d**: NO), the lock control unit **52** repeats step **S140d**. On the other hand, when it is specified that the current mechanical angle of the motor **11** has reached the reference angle (step **S140d**: YES), the lock control unit **52** determines whether or not there is a lock command from the flight control system **40** (step **S170**). If it is determined that there is no lock command (step **S170**: NO), the process returns to step **S114d**. On the other hand, when it is determined that there is the lock command (step **S170**: YES), the rotation of the rotary blade **30** is locked by locking the rotation of the motor **11** (step **S180**). In step **S180**, the rotation of the motor **11** is locked as a target when the current mechanical angle of the motor **11** reaches the stored reference angle and the rotation of the rotary blade **30** is locked.

(80) According to the control device **50** of the EDS **10** in the fifth embodiment described above, the same effect as that in the third embodiment is obtained. In addition, the angle sensor **13** for detecting the mechanical angle of the motor **11** is provided in the motor **11**, and the rotation of the rotary blade **30** is locked as a target when the current mechanical angle of the motor **11** reaches the stored reference angle. Therefore, it is possible to omit the calculation of the machine angle.

F. Sixth Embodiment

(81) In the EDS **10** on which the control device **50** of the sixth embodiment is mounted, the output of the motor **11** is transmitted to the rotary blade **30** via the gear **G**, the rotary blade **30** is provided with an angle sensor so that the angle acquisition unit **53** acquires the rotation angle of the rotary blade **30**, and the reference angle storage unit **54** store the reference angle, which is an angle corresponding to the target lock position, as the value of the rotation angle of the rotary blade **30**. In this respect, the EDS**10** of the sixth embodiment is different from the EDS**10** of the fifth embodiment. Along with this difference, the lock processing of the sixth embodiment is different from the lock processing of the fifth embodiment in that steps **S110e** and **S140e** are executed instead of steps **S110** and **S140d**. Since the other configurations are the same as those in the fifth embodiment, the same configurations are designated by the same reference numerals, and detailed description thereof will be omitted.

(82) In step **S110e** shown in FIG. **15**, the lock control unit **52** detects a reference angle, that is, an angle corresponding to a target lock position when the rotary blade **30** is locked in the lock angle range (step **S110**). In the present embodiment, the reference angle is stored in the reference angle storage unit **54** as the value of the rotation angle of the rotary blade **30** when the rotation of the rotary blade **30** is locked last time. Such a reference angle may be stored in the reference angle storage unit **54** as a value of the rotation angle of the rotary blade **30** when the start switch of the eVTOL **100** is turned on.

(83) The lock control unit **52** specifies whether or not the current rotation angle of the rotary blade **30** has reached the reference angle (step **S140e**). The current rotation angle of the rotary blade **30** is measured by the angle sensor provided in the motor **11** and acquired by the angle acquisition unit **53**. When it is specified that the current rotation angle of the rotary blade **30** has not reached the reference angle (step **S140e**: NO), the lock control unit **52** repeats step **S140e**. On the other hand, when it is specified that the current rotation angle of the rotary blade **30** has reached the reference angle (step **S140e**: YES), the lock control unit **52** determines whether or not there is a lock command from the flight control system **40** (step **S170**). If it is determined that there is no lock command (step **S170**: NO), the process returns to step **S114e**. On the other hand, when it is determined that there is the lock command (step **S170**: YES), the rotation of the rotary blade **30** is locked by locking the rotation of the motor **11** (step **S180**). In step **S180**, the rotation of the motor **11** is locked as a target when the current rotation angle of the rotary blade **30** reaches the stored reference angle and the rotation of the rotary blade **30** is locked.

(84) According to the control device **50** of the EDS **10** in the sixth embodiment described above, the same effect as that in the fifth embodiment is obtained. In addition, the angle sensor for

detecting the rotation angle of the rotary blade **30** is provided on the rotary blade **30**, and the rotation of the rotary blade **30** is locked as a target when the current rotation angle of the rotary blade **30** reaches the stored reference angle. Therefore, even in a configuration in which the output of the motor **11** is transmitted to the rotary blade **30** via the gear **G**, the rotation of the rotary blade **30** can be locked within the lock angle range.

G. Seventh Embodiment

(85) As shown in FIG. **16**, the EDS **10** of a seventh embodiment is mounted on the eVTOL **100f**. The eVTOL **100f** is different from the eVTOL **100** of the first embodiment in that it further has a detection unit for detecting the rotational position of the rotary blade **30**. Along with this difference, the lock processing of the seventh embodiment shown in FIG. **17** is different from the lock processing of the first embodiment in that steps **S110** to **S130** and **S160** are omitted and step **S140f** is executed instead of step **S140**. Since the other configurations are the same as those in the first embodiment, the same configurations are designated by the same reference numerals, and detailed description thereof will be omitted. In the present embodiment, the output of the motor **11** is transmitted to the rotary blade **30** without the gear, but may be transmitted to the rotary blade **30** via the gear.

(86) The eVTOL **100f** of the seventh embodiment is provided with a visual servo **92** and an infrared sensor **94** as a detection unit for detecting the rotational position of the rotary blade **30**. The visual servo **92** is composed of, for example, a camera provided on the tail wing **28**, and detects the rotational position of each rotary blade **30**. The infrared sensor **94** is provided, for example, vertically below each rotary blade **30**, and detects the rotational position of each rotary blade **30**. Either one of the visual servo **92** and the infrared sensor **94** may be omitted, and instead of the infrared sensor **94** or in addition to the infrared sensor **94**, any detection unit capable of detecting the rotational position of the rotary blade **30** such as an ultrasonic sensor, a position sensor, an optical sensor, a photoelectric sensor, or the like may be provided.

(87) As shown in FIG. **18**, in the pulse output unit **55** included in the EDS **10** of the seventh embodiment, based on the detection results of the visual servo **92** and the infrared sensor **94**, a pulse is output when the current rotational position of the rotary blade **30** coincides with the target lock position for locking the rotary blade **30** within the lock angle range. With such a configuration, the pulse output unit **55** outputs a pulse every time the rotary blade **30** makes one rotation.

(88) As shown in FIG. **17**, the lock control unit **52** specifies whether or not the current rotational position of the rotary blade **30** detected by the visual servo **92** and the infrared sensor **94** has reached the target lock position (step **S140f**). When it is specified that the current rotational position of the rotary blade **30** has not reached the target lock position (step **S140f**: NO), the lock control unit **52** repeats step **S140f**. On the other hand, when it is specified that the current rotational position of the rotary blade **30** has reached the target lock position (step **S140f**: YES), the lock control unit **52** determines whether or not there is a lock command from the flight control system **40** (step **S170**). If it is determined that there is no lock command (step **S170**: NO), the process returns to step **S114f**. On the other hand, when it is determined that there is the lock command (step **S170**: YES), the rotation of the rotary blade **30** is locked by locking the rotation of the motor **11** (step **S180**). In step **S180**, the lock control unit **52** locks the rotary blade **30** as the target when the pulse output unit **55** outputs a pulse. Execution of step **S180** ends the lock processing.

(89) According to the control device **50** of the EDS **10** in the seventh embodiment described above, the same effect as that in the first embodiment is obtained. In addition, the detection unit that detects the rotational position of the rotary blade **30** is provided in the eVTOL **100f**, and the pulse output unit **55** outputs the pulse when the target lock position for locking the rotary blade **30** within the lock angle range coincides with the current rotational position of the rotary blade **30**. Therefore, the detection of the electric angle, the calculation of the mechanical angle, and the like can be omitted, and the reference angle storage unit **54** can be omitted. In addition, even in a configuration in which the output of the motor **11** is transmitted to the rotary blade **30** via the gear **G**, the rotation

of the rotary blade **30** can be locked within the lock angle range.

H. Other Embodiments

H-1. Other Embodiment 1

(90) In each of the above embodiments, each rotary blade **30** has two blades **33**, but the rotary blades **30** are not limited to two, and may have an arbitrary number of blades **33** such as one or three. In a configuration having one blade **33**, when the number of blades **33** of the rotary blade **30** is A and the target lock position is B, the lock angle range may be set to an angle range of [B-45 deg] or more and [B+45 deg] or less. According to such an embodiment, it is possible to further suppress that the angle at which the rotary blade **30** is locked deviates significantly from the target lock position.

H-2. Other Embodiment 2

(91) In each of the above embodiments, the control device **50** is mounted on each EDS **10**, but even if one control device **50** of the EDS **10** is provided on the eVTOL **100** and **100f** and the control device **50** controls a plurality of EDSs **10**. In such a configuration, the lock control unit **52** may specify the rotary blade **30** to be locked among the plurality of rotary blades **30** according to the operation modes of the eVTOL **100** and **100f**. For example, when the operation mode of the eVTOL **100**, **100f** is the cruising mode, the lift rotary blade **31** may be specified as the locking rotary blade **30**. According to such an embodiment, it is possible to appropriately lock the rotary blade **30** which is not rotationally driven among the plurality of rotary blades **30** according to the operation mode of the eVTOL **100**, **100f**.

H-3. Other Embodiment 3

(92) The EDS **10** of each of the above embodiments has a motor **11** for rotationally driving the rotary blade **30**, but the EDS **10** may not include the motor **11**. That is, the EDS **10** may be a system for controlling the motor **11** provided outside the EDS **10**. Even with such a configuration, the same effect as that of each of the above embodiments can be obtained.

H-4. Other Embodiment 4

(93) The configurations of eVTOL **100**, **100f** in each of the above embodiments are merely examples and can be changed in various ways. For example, in each of the above embodiments, the eVTOL **100**, **100f** includes the rotor **32** having a variable tilt angle, but instead of the tilt rotor **32** or in addition to the tilt rotor **32**, a cruise rotor with a fixed tilt angle may be provided as the rotary blade **30**. Further, for example, the tilt rotor **32** may be provided with an accommodating portion, and the tilt rotor **32** that is not rotationally driven may be accommodated in the accommodating portion. In such an embodiment, the tilt rotor **32** may be stopped and locked at an angle that allows it to be accommodated in the accommodating portion. Further, for example, the accommodating portion of the lift rotary blade **31** may be omitted. In such an embodiment, the lift rotary blade **31** may be stopped and locked at an angle such that the value of air resistance becomes small. Even with such a configuration, the same effect as that of each of the above described embodiments can be obtained.

H-5. Other Embodiment 5

(94) The EDS **10** of each of the above embodiments has been mounted on the eVTOL **100**, **100f**, but is not limited to be mounted on the eVTOL **100**, **100f** and may be mounted on any manned or unmanned electric aircraft. It is not limited to be mounted on the electric aircraft, and may be mounted on an electric moving body such as a ship or a submersible that is remotely controlled by radio, or any ship that is rotationally driven by a screw. That is, in general, the control device **50** of the EDS **10** may be configured to rotationally drive one of the rotary blades **30** and the screw that are the rotating body. Further, the lock angle range may be an angle range in which the resistance value is smaller than the intermediate value between the maximum value and the minimum value of the resistance value that the rotating body receives from the external fluid according to the lock position of the rotating body. Even with such a configuration, the same effect as that of each of the above embodiments can be obtained.

H-6. Other Embodiment 6

(95) In the eVTOL **100** of the first to sixth embodiments, an arbitrary detection unit capable of detecting the rotational position of the rotary blade **30** may be provided in the eVTOL **100f** as in the case of the eVTOL **100f** of the seventh embodiment, and the rotary blade **30** may be locked at the target lock position by cooperating with such a detection unit. According to such a configuration, the accuracy when locking the rotary blade **30** in the lock angle range can be improved.

(96) The present disclosure should not be limited to the embodiments described above, and various other embodiments may be implemented without departing from the scope of the present disclosure. For example, the technical features in each embodiment corresponding to the technical features in the form described in the summary may be used to solve some or all of the above-described problems, or to provide one of the above-described effects. In order to achieve a part or all, replacement or combination can be appropriately performed. Also, if the technical features are not described as essential in the present specification, they can be deleted as appropriate.

(97) The control device, the external device, and the technique according to the present disclosure may be achieved by a dedicated computer provided by constituting a processor and a memory programmed to execute one or more functions embodied by a computer program. Alternatively, the controller and the method described in the present disclosure may be implemented by a special purpose computer configured as a processor with one or more special purpose hardware logic circuits. Alternatively, the control device, the external device, and the technique according to the present disclosure may be achieved using one or more dedicated computers constituted by a combination of the processor and the memory programmed to execute one or more functions and the processor with one or more hardware logic circuits. The computer programs may be stored, as instructions to be executed by a computer, in a tangible non-transitory computer-readable medium.

Claims

1. An electric vertical take-off and landing aircraft comprising: an airframe; a plurality of rotary blades; a plurality of electric drive systems respectively corresponding to the plurality of rotary blades for rotationally driving each of the plurality of rotary blades, a subset of the plurality of electric drive systems rotating and driving at least one lift rotary blade for obtaining lift of the airframe, and another subset of the plurality of electric drive systems rotating at least one tilt rotor with a variable tilt angle; and a flight control system, wherein: each electric drive system of the plurality of electric drive systems includes: an inverter circuit; a motor that (i) outputs torque corresponding to voltage and current supplied from the inverter circuit and (ii) drives the rotary blade corresponding to the electric drive system to rotate via a shaft; and a control device; the flight control system is configured to output, to the control device of each of the plurality of electric drive systems, a lock command for stopping and locking rotation of a rotary blade that is not driven to rotate, the lock command being output to the control device of the electric drive system corresponding to the lift rotary blade when an operation mode of the electric vertical take-off and landing aircraft is a first mode, and the lock command being output to the control device of the electric drive system corresponding to the tilt rotor when an operation mode of the electric vertical take-off and landing aircraft is a second mode; and the control device of each electric drive system includes: a processor programmed to: acquire a rotation angle of the corresponding rotary blade; and perform control to lock the corresponding motor so as to lock a rotation of the corresponding rotary blade within a set lock angle range based on the acquired rotation angle by gradually reducing a rotation speed of the corresponding rotary blade and locking the rotation of the corresponding rotary blade after the rotation speed becomes equal to or less than a predetermined rotation speed; and a memory configured to store a reference angle for locking the corresponding rotary blade within the lock angle range as (i) a value of an electric angle of the corresponding

motor when the rotation of the corresponding rotary blade was last locked or (ii) as a value of the electric angle of the corresponding motor when a start switch of the electric vertical take-off and landing aircraft is turned on.

2. The electric vertical take-off and landing aircraft according to claim 1, wherein the lock angle range is an angle range in which a resistance value is smaller than an intermediate value between a maximum value and a minimum value of the resistance value that the corresponding rotary blade receives from an external fluid according to a lock position of the corresponding rotary blade.

3. The electric vertical take-off and landing aircraft according to claim 1, wherein the plurality of electric drive systems are mounted on the airframe of the aircraft, the lift rotary blade is housed in an accommodating portion provided in the airframe in a state where the lift rotary blade is not rotationally driven, and the lock angle range is included in an accommodation angle range, which is an angle range in which the lift rotary blade can be housed in the accommodating portion.

4. The electric vertical take-off and landing aircraft according to claim 1, wherein each of the plurality of rotary blades has blades formed symmetrically about a rotation axis along a direction perpendicular to the rotation axis, and the lock angle range is $[B-2\pi/A]$ or more $[B+2\pi/A]$ or less when a number of blades is A, a target lock position is B, and 180 deg is π .

5. An electric vertical take-off and landing aircraft comprising: an airframe; a plurality of rotary blades; a plurality of electric drive systems respectively corresponding to the plurality of rotary blades for rotationally driving each of the plurality of rotary blades, a subset of the plurality of electric drive systems rotating and driving at least one lift rotary blade for obtaining lift of the airframe, and another subset of the plurality of electric drive systems rotating at least one tilt rotor with a variable tilt angle; and a flight control system, wherein: each electric drive system of the plurality of electric drive systems includes: an inverter circuit; a motor that (i) outputs torque corresponding to voltage and current supplied from the inverter circuit and (ii) drives the rotary blade corresponding to the electric drive system to rotate via a shaft; and a control device; the flight control system is configured to output, to the control device of each of the plurality of electric drive systems, a lock command for stopping and locking rotation of a rotary blade that is not driven to rotate, the lock command being output to the control device of the electric drive system corresponding to the lift rotary blade when an operation mode of the electric vertical take-off and landing aircraft is a first mode, and the lock command being output to the control device of the electric drive system corresponding to the tilt rotor when an operation mode of the electric vertical take-off and landing aircraft is a second mode; and the control device of each electric drive system includes: a processor programmed to: acquire a rotation angle of the corresponding rotary blade, which is provided with an angle sensor that detects the rotation angle; perform control to lock the corresponding motor so as to lock a rotation of the corresponding rotary blade within a set lock angle range based on the acquired rotation angle by gradually reducing a rotation speed of the corresponding rotary blade and locking the rotation of the corresponding rotary blade after the rotation speed becomes equal to or less than a predetermined rotation speed; and perform control to lock the corresponding rotary blade as a target when a current rotation angle of the corresponding rotary blade reaches a stored reference angle, the reference angle being stored (i) when the rotation of the rotary blade was last locked or (ii) when a start switch of the electric vertical take-off and landing aircraft is turned on; and a memory configured to store the reference angle for locking the corresponding rotary blade within the lock angle range as a value of an electric angle of the corresponding motor.
